Title

Electrophysiological correlates underlying the use of generated affective music to regulate stress in individuals with elevated anxiety

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Abstract

Recent advances in generative music systems offer new opportunities for supporting stress regulation in individuals experiencing psychological distress. However, the degree to which music should reflect the desired relaxation state, and the neural dynamics underlying music-based regulation, are not well understood. The present study investigated the electrophysiological correlates of stress reduction using generated music that expresses different degrees of relaxation. Twenty-seven adults (10M, age: 31.33 ± 9.65 years) with elevated anxiety (measured by the Depression, Anxiety, Stress Scale-21) completed three blocks of a stress-inducing arithmetic task, each followed by three minutes of silence, Low-relaxation music (Low-RM), or High-relaxation music (High-RM). Low-RM anchored musical characteristics near participants' post-stress affect ratings; High-RM shifted these parameters 30% toward positive-valence and low-arousal. Following stress induction, subjective valence increased in both music conditions but arousal decreased only in High-RM and silence. Relative to silence, frontal delta suppression was observed in both music conditions, with additional parietal delta suppression in Low-RM. Exploratory analyses identified parieto-occipital alpha suppression during High-RM. Across conditions, greater frontal delta reduction was associated with lower Anxiety scores. These findings provide new insights for the efficacy of affective generated music in modulating neurophysiological processes implicated in stress regulation for individuals with elevated anxiety.

1. Introduction

Mental health conditions constitute a substantial and growing global health burden. According to the World Health Organization (WHO), approximately one in seven people globally live with a mental disorder, with anxiety and depressive disorders representing the most prevalent conditions [1]. Anxiety and depression frequently co-occur, with comorbidity reported in up to 75% of cases [2], although anxiety often precedes depression or constitutes the primary condition [3].

Anxiety is characterized by heightened vigilance, increased attentional bias toward threat [4-6], and persistent difficulties in down-regulating negative affect [7]. These regulatory difficulties in individuals with elevated anxiety are not merely behavioral, but are accompanied by measurable disruptions in physiological markers such as heart rate variability [8,9] and neural oscillatory dynamics [10-12]. In particular, studies have commonly reported increased frontal delta [13], disrupted frontal alpha asymmetry during resting-state electroencephalography (EEG) - with left frontal hypoactivation observed in depressed patients [14-20], and altered lateralization of parietotemporal alpha activity to be associated with elevated anxiety levels [19] and depression [15].

Consequently, these behavioral attributes and physiological dysfunction in individuals with elevated anxiety may increase their susceptibility to the effects of stressors. Chronic stress further compounds these difficulties, driving adverse immune and physiological outcomes including dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis, increased inflammatory markers, and reduced resistance to illness [21-24]. Despite the substantial societal burden and debilitating consequences imposed by these mental health conditions, access to effective and scalable interventions remains limited, and the gap between the prevalence of psychological distress and treatment uptake continues to widen [25]. It is therefore a pressing public health priority to develop accessible strategies to support stress recovery in individuals with psychological distress.

In this regard, music-based intervention presents as a promising non-pharmacological solution which has been shown to reduce stress [26-28] and anxiety [29] in the general population, as well as those with elevated anxiety [30, 31]. The efficacy of using music for reducing stress or anxiety is often attributed to the ability of relaxing music to reduce cortisol levels [32], modulate the HPA axis activity [33] and patterns of brain activity that are dysregulated in individuals with clinical depression [34] and non-clinical anxiety [35].

Furthermore, music listening engages neural networks that substantially overlap with those dysregulated in these affective disorders. Music listening recruits distributed brain networks encompassing auditory processing, emotional regulation, memory systems, and reward pathways [36, 37], with the prefrontal cortex-hippocampus-amygdala circuit playing a central role in modulating stress-related emotions, including anxiety and fear [38]. Studies have found that music listening produces changes in oscillatory power across frontal regions across delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz) frequency bands [39-41]. Beyond frontal regions, studies have increasingly reported additional involvements of central, temporal, and parietal

regions [41-43], which could be a result of music listening involving cognitive processes such as attention, working memory, and predictive processing [44, 45].

Recovery from a stressor is generally more effective when participants listened to music which is low in arousal and positive in valence [46]. However, the majority of music-based stress research has treated music as a homogeneous intervention, with relatively little attention paid to the critical question of *how relaxing* the music should be relative to the listener's current emotional state. This question is addressed by the **iso principle**, a foundational concept in music therapy. Originating in 1940s, the iso principle involves initially matching the music to the listener's current affective state and then gradually shifting toward a desired emotional state can be more effective for mood regulation as compared to listening to music that outright targets the desired state [47]. Recent evidence suggests that meeting listeners "where they are" emotionally before attempting to shift their mood is an important component of successful mood modulation [48], and that affect-sequencing can be effectively applied through systematically designed musical playlists [49, 50]. This is further supported by the **mood congruence hypothesis** [51], which holds that listeners engage more readily and deeply with music that matches their current affective state [52]. Beyond mood congruence, listening to music of a preferred genre has also shown to be more effective in inducing targeted emotions [53].

However, findings from the relatively fewer studies testing the iso principle in non-clinical contexts have been less consistent. For instance, a recent study [54] found that among healthy young adults, there was no difference in anxiety reduction between music playlists adopting the iso principle and those using the compensatory principle (i.e., the proposition that music with the desired affective qualities, contrasting with one's current feelings, will be effective in modulating emotion; [55]). Further, individual factors including personality [56], and the type of mood regulatory strategies an individual uses [57], have been shown to influence the efficacy of music-based emotion regulation. One plausible explanation for these inconsistencies is that naturalistic playlist-based paradigms cannot precisely control the acoustic and affective properties of stimuli. As such, the degree to which music actually diverges from a listener's current emotional state may likely be confounded with other non-musical variables, making it difficult to isolate the contribution of affective matching per se. This constraint can now be addressed through recent advances in generative music systems, which enable the production of music anchored to pre-specified levels of valence and arousal in real-time [58]. In particular, AffectMachine-Classical [59] and AffectMachine-Pop [60] are generative music systems capable of creating pleasant, unfamiliar instrumental music where one can explicitly manipulate the level of arousal and valence to test a hypothesis of interest.

The present study thus aimed to investigate the electrophysiological correlates of stress recovery using generated affective music which differed in the degree of relaxation content in a sample of adults with elevated anxiety. Rather than strictly implementing the iso principle (i.e., designing the music to move gradually towards a state of relaxation), we compared the impact of music with lower degree of relaxation (Low-Relaxation Music; Low-RM) versus a higher degree of relaxation (High-Relaxation Music; High-RM) in the valence-arousal space. This design was

chosen based on two reasons: (1) to enable clear behavioral and neural comparisons between these affective states, and (2) to overcome the difficulty of distinguishing continuous neural changes underlying the affective trajectory change inherent in iso-principle designs. In the Low-RM condition, musical parameters were anchored to the participant's self-reported post-stress valence and arousal levels, whereas these parameters were shifted toward a calmer, more positive affective target in the High-RM. EEG was recorded continuously as participants completed three blocks of a stress-inducing mental arithmetic task, each followed by a three-minute recovery period of silence, Low-RM, or High-RM. Participants also completed behavioral questionnaires measuring personality, music-based regulation strategies, music preferences and experience. By combining behavioral and electrophysiological measures, this study aims to provide a more holistic understanding of how the degree of relaxation embedded in generative music shapes the neural and affective processes involved in recovering from stress in a sample of individuals with elevated anxiety, after accounting for the listener's choice of preferred music genre.

2. Methods

2.1. Participants

Participants were recruited via advertisements placed around Nanyang Technological University (NTU), Singapore, and through social media. The study was advertised as an investigation of the neural correlates of mathematics and music to avoid priming participants to the stress-regulation aim. Ethical approval was granted by NTU Institutional Review Board (No. IRB-2024-434), and all methods were performed in accordance with the relevant institutional guidelines and regulations.

Individuals were excluded if they self-reported having any diagnosed neurological conditions, currently using psychotropic medication, or have uncorrected hearing or vision impairment. Recruitment targeted individuals scoring in the elevated range on the Stress subscale of the Depression, Anxiety, Stress Scale-21 [61], and was subsequently broadened to include individuals with elevated scores on any of the DASS-21 subscales due to initial recruitment constraints.

A total of 39 adults (14M, age range: 21-54 years, $M = 32.03$, $SD = 10.97$) were recruited, and informed consent was obtained from all participants prior to the start of experiment. Following exclusion for technical issues ($n = 6$), missing data ($n = 1$) and poor EEG data quality ($n = 4$), 28 participants remained. As all but one participant scored Severe-to-Extremely-Severe on the Anxiety subscale, the one participant scoring in the Moderate range was excluded in the final analyses to establish sample homogeneity with respect to anxiety severity. This exclusion was made on the basis of inspection of distribution of DASS-21 scores alone, prior to any EEG or behavioral analyses. The final sample for analyses thus comprised 27 participants (10M, age range: 21-53 years, $M = 31.33$, $SD = 9.65$).

2.2. Procedure

The experiment was programmed and presented with PsychoPy3 (v2024.2.3) [62]. Participants first completed questionnaires related to music-based mood regulation; personality; depression, anxiety, and stress; music preferences and experience (see Section 2.3.3 for more details). Next, they were fitted with the EEG cap. Participants sat in a dimly lit room and were presented with verbal as well as written instructions on the computer screen for the experiment. In particular, they were first introduced to the definitions of *valence* and *arousal*, and the experimenter ensured that participants understood these terms correctly. The audio stimuli were played through loudspeakers (AudioEngine HD3), and participants were asked to adjust the volume to a comfortable level and maintain that volume for the duration of the experiment. Participants completed three blocks of a stress-inducing mental arithmetic task, which is each followed by a 3-minute recovery period of silence, Low-relaxation music (Low-RM), or High-relaxation music (High-RM). In the Low-RM condition, musical parameters were anchored to the participant's self-reported post-stress valence and arousal levels, whereas these parameters were shifted 30% toward positive-valence and low-arousal relative to the Low-RM in the High-RM condition. The music was generated by either AffectMachine-Classical [59] or AffectMachine-Pop [60], depending on their indication for pop or classical music.

For each block, participants provided arousal-valence (A-V) ratings at three time points: before the arithmetic task, after the arithmetic task, and after the silence- or music-listening period. Both arousal and valence were rated on a scale of 0 (*arousal: Less Energy; valence: Less Positive*) to 10 (*arousal: More Energy; valence: More Positive*), with 5 being neutral. In addition, discrete emotions were measured after the music listening conditions using the Geneva Music-Induced Affect Checklist (GEMIAC) [63]. It consists of 14 terms: wonder, moved, enchanted, inspired, energetic, joyful, powerful, tender, relaxed, melancholy, sentimental, indifferent, tense, and agitated. Each item was rated on a scale of 1 (*not at all*) to 5 (*very much*). After the last block, participants answered questions related to the arithmetic task and their mathematical abilities with a continuous slider from 0 (*not at all*) to 100 (*agree completely*): "I found the questions easy", "I found the task stressful", "I wish I had more time to answer the questions", "In general I am good at maths", "I am happy with my performance in the task" (Fig. 1).

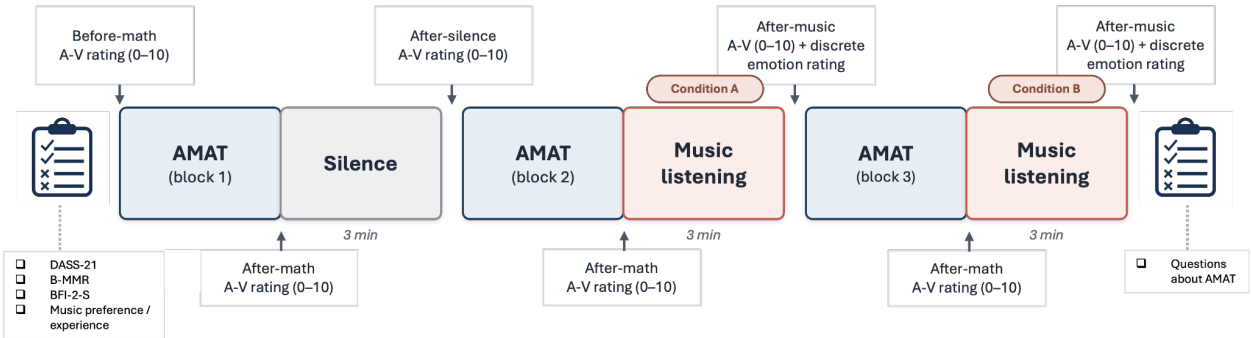


Figure 1. Overview of the experimental paradigm whereby stress is induced through the Adaptive Mental Arithmetic Task (AMAT), and is followed by silence or music listening conditions. The

music listening conditions are counterbalanced. *Notes.* A-V: Arousal-Valence. B-MMR: Brief Music in Mood Regulation Scale. DASS-21: Depression, Anxiety and Stress Scale 21. BFI-2-S: Short form of the Big Five Inventory-2

2.3. Materials

2.3.1. Stress induction task

In this study, we developed an Adaptive Mental Arithmetic Task (AMAT), modified from the repeated Montreal Imaging Stress Test (rMIST) [64] to induce stress. While the rMIST represents an advance over single-session stress paradigms in its ability to elicit consistent self-reported and physiological reactivity across repeated exposures, this approach is often limited by constraints in intra-session stimulus novelty. This constraint may attenuate the experienced stress over time, thereby producing a habituation effect. Furthermore, the original MIST incorporated a failure algorithm that adjusted time limits based on individual performance during a pre-task training phase. This pre-specification of difficulty may limit the extent to which the task remains individually challenging and stressful throughout, particularly for participants whose arithmetic performance deviates substantially from the group mean. The AMAT was therefore designed to address these limitations by introducing trial-level adaptive difficulty and greater stimulus variability across blocks.

The AMAT was developed as a computer-based arithmetic task that dynamically adjusted task difficulty based on the individual's performance throughout the duration of the task. This was implemented by (1) increasing the value of numbers, (2) using division and multiplication rather than addition and subtraction, or (3) combining multiple operations in an equation. Participants were instructed to provide their answers using the arrow keys "Top", "Bottom", "Left", "Right" instead of typing their answers. This was designed to minimize motion artefact in the EEG recording.

To increase stress and mitigate potential habituation effects across blocks, several strategies were used: (1) *Adaptive difficulty*: the task increased in difficulty when participants were performing well, to guarantee all participants achieved a similar failure rate, (2) *Social comparative stress*: feedback was shown to participants on screen, telling them that they are performing poorly compared to their peers regardless of actual performance, (3) *Incorrect response stress*: participants received real-time feedback when an incorrect answer is given, in the form of a red 'X' and a buzzer sound, and (4) *Financial incentive*: participants are informed that their performance in the task will affect the financial compensation, such that incorrect answer lowers their financial reward. This was deception as all participants received the full SGD $100 in vouchers at the end of the study.

2.3.2. Music generation systems: AffectMachine

In this study, 13 participants listened to AffectMachine-Classical while 14 participants listened to AffectMachine-Pop, as informed by their musical preference. Both AffectMachine systems were developed in Python and generate music through rule-based compositional frameworks that manipulate musical features shown to influence emotional expression in music,

such as harmonic, rhythmic, timbral, and loudness parameters. Each system was designed to follow the stylistic conventions of their respective genre (for more details, see [59, 60]). The systems accept a sequence of arousal-valence (A-V) coordinate pairs as input, each within the normalized range [0, 1].

The systems take A-V values of the self-reported ratings after the mental arithmetic task as the first value to generate music. Corresponding harmonic, rhythmic, and timbral parameters are then encoded in the form of a MIDI event stream as output. These musical parameters are updated in real time at each bar in accordance with the current A-V values. The resulting MIDI event stream is transmitted to Ableton digital audio workstation (DAW) over virtual MIDI buses, where it is rendered into audio (Fig. 2a). Both systems have been validated through listening studies, which demonstrated that listeners reliably perceived the generated music as expressing the intended target emotions [59, 60].

Music Stimuli. For both conditions, a machine learning approach was used to map the continuous EEG signal to A-V values in real-time, with the A-V values serving as input parameters for AffectMachine. Specifically, EEGNet [65] was deployed to generate output for arousal and valence separately every four seconds. EEGNet is a compact convolutional neural network developed for EEG-based brain-computer interfaces, using depthwise and separable convolutions to extract temporal and spatial features from EEG signals. It is robust across different task paradigms and can be trained with limited data [65].

The output of the A-V models were within a narrow range for both conditions: A-V values for the Low-relaxation music (Low-RM) condition were closer to the self-reported emotion ratings post-math task as compared to the High-relaxation music (High-RM) condition (**Low-RM:** Arousal = 2.27 ± 1.81, Valence = 2.60 ± 1.77; **High-RM:** Arousal = 3.79 ± 1.91 and Valence = 4.50 ± 2.47). Consequently, the music in the Low-RM condition comprised significantly different A-V values compared to the High-RM condition at $p < .05$ (Fig. 2b, c).

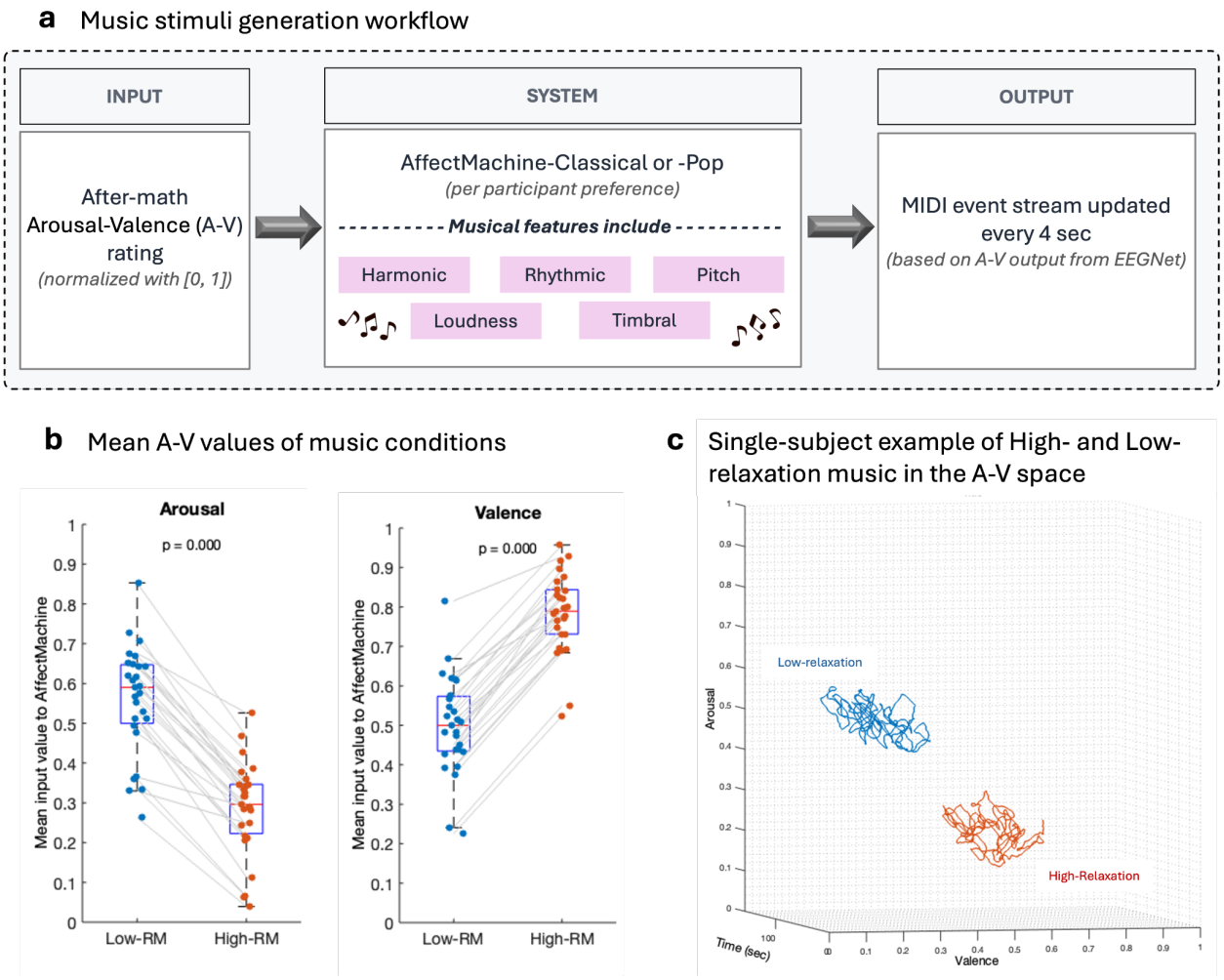


Fig 2. (a) Pipeline of AffectMachine music generation system. The system takes as input the self-reported arousal-valence ratings as a starting point to generate music, and the A-V coordinate pair is updated every four seconds based on the mapping between EEG signal and A-V state via EEGNet. (b) High-relaxation music generated is significantly lower in arousal and higher in valence as compared to Low-relaxation music. (c). An example of High- and Low-relaxation music generated for one participant in the arousal-valence space. *Notes.* Low-RM: Low-relaxation music. High-RM: High-relaxation music. A-V: arousal-valence

2.3.3. Questionnaires for individual differences

Participants completed the Depression, Anxiety and Stress Scale 21 (DASS-21) [61] as a measure of their trait-level depression, anxiety, and stress. Motivated by literature that suggests associations between mood regulation strategies and personality with music-based mood regulation [56, 57], participants also completed relevant validated questionnaires: Music-based mood regulation was measured by the Brief Music in Mood Regulation Scale (B-MMR) [66]; personality was measured by the Short form of the Big Five Inventory-2 (BFI-2-S) [67], which assesses the Big Five personality domains and 15 more specific facet traits. We limit the current

analyses to the five domains, as it is recommended that facet-level traits can only be useful for examination in reasonably large samples (approximately ≥ 400 observations) [67]. Lastly, participants completed questions related to musical taste and experience at the end of the experimental block (see Supplementary Table S1 online for further details of the questionnaires).

2.4. EEG acquisition

EEG data were recorded using a 64-channel gel-based actiCAP (Brain Products, Gilching, Germany) arranged according to the extended 10-20 system. The sampling rate was set at 1000 Hz. The ground electrode was placed at the centre of the forehead, and the reference was placed at FCz. Two scalp electrodes (PO7, PO8) were repurposed to measure electrooculography (EOG): one positioned at the lateral canthus of the right eye, and one positioned infraorbitally to the right eye. Consequently, 61 scalp electrodes were available for subsequent analyses.

2.5. EEG preprocessing

EEG data were preprocessed using a custom automated pipeline implemented in Matlab R2019b with EEGLAB [68]. Continuous EEG data were first bandpass filtered between 1 and 40 Hz using EEGlab's finite impulse response filtering function, then downsampled to 250 Hz. Next, EOG channels were identified and extracted for subsequent eye-blink artifact correction. Automatic bad channel detection was performed on non-EOG channels using a kurtosis-based criterion with a z-score threshold of ± 5. Channels identified as noisy were interpolated using spherical spline interpolation to preserve scalp topography. On average, 3 channels were interpolated per participant.

Eye-movement artifacts were corrected using a regression-based EOG correction approach [69]. EOG auto-covariance and EEG-EOG cross-covariance matrices were computed across the continuous recording to derive global ordinary least-square regression weights, estimating the contribution of ocular activity to each EEG channel. The weighted EOG signal was subsequently subtracted from the EEG to obtain corrected data, after which EOG channels were removed, retaining 61 scalp channels. The continuous corrected EEG data without EOG channels were then segmented into three epochs of interest corresponding to condition onsets: Silence, High-RM, and Low-RM. Each epoch spanned -10 s to +180 s relative to condition onset, yielding a total epoch length of 190s. The 10-sec pre-onset buffer was retained to provide a baseline window for subsequent time-frequency analysis and to guard against wavelet edge effects at low frequencies.

To remove residual non-ocular artifacts, Independent Component Analysis (ICA) was performed using the Extended Infomax algorithm implemented in EEGLab (runica, extended mode). Prior to ICA decomposition, data rank was computed accounting for the number of interpolated channels, and was passed as the argument for Principal Component Analysis (PCA) dimension reduction to ensure a full-rank decomposition. Importantly, common average referencing was deferred until after ICA to avoid an additional rank reduction during decomposition. Artifactual ICs were then identified and removed automatically using the ADJUST toolbox [70], which classifies components into eye blinks, horizontal eye movements, generic

discontinuities, and/or abnormal spectral profiles. On average, 13 ICs were removed per participant. Finally, the ICA-cleaned data were re-referenced to the common average reference (CAR) [71] computed across all 61 scalp channels, excluding any previously interpolated channels from the reference computation.

2.6. EEG analysis: Time-frequency decomposition and statistical analyses

A time-frequency representation (TFR) was derived for each condition by computing complex Morlet wavelet transform (CWT) for each channel and condition. The Morlet wavelet was chosen as it is the most established type of wavelet for EEG signals and thus recommended [72]. The wavelet kernel length was set to 6 seconds. The frequency range was linearly spaced and set to 2 - 30 Hz with a resolution of 1 Hz to cover the most relevant frequencies. The number of wavelet cycles was set to gradually increase from 10 cycles for the lowest frequency to 30 cycles for the highest frequency. For each cleaned epoch, baseline correction was performed by subtracting the average amplitude of the period -5 to 0 sec from the entire signal epoch. Lastly, we excluded the first and last 10-sec of each epoch for analyses as these were time segments subjected to more movement as a result of participants moving around to respond to questions.

2.6.1. *A priori* ROI analyses and associations with questionnaires for individual differences

Region-of-interest (ROI) analyses were conducted across frontal (F1, F2, F3, F4, F5, F6), central (C1, C2, C3, C4, C5, C6), and parietal (P1, P2, P3, P4, P5, P6) regions, motivated by existing literature implicating these regions in delta (2-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz) frequency bands. For each ROI, time-frequency power was averaged across the constituent channels at the single-subject level prior to any statistical analyses.

Prior to statistical testing, temporal pre-processing was applied to each epoch. As the conditions of interest involved sustained auditory stimulation spanning 170 seconds rather than discrete time-locked events, fine temporal resolution was not expected to be informative and was likely to introduce noise into group-level comparisons. To address this, a Gaussian smoothing kernel of 10-sec was first applied along the time dimension of each participant's time-frequency map, followed by aggregation into non-overlapping 5-second time bins by averaging within each bin. Normality of the difference between conditions was next assessed at each time-frequency bin using the Shapiro-Wilk test. The proportion of cells consistent with normality was reported as a guide to test selection: if one or more group-level comparison showed that more than 50% of time-frequency cells violated normality assumptions, the Wilcoxon signed-rank test was used instead of paired *t*-test for all comparisons.

Group-level comparisons between condition pairs (i.e., High-RM vs. Silence, Low-RM vs. Silence, High-RM vs. Low-RM) were conducted using a cluster-based permutation procedure [73] applied separately per frequency band and ROI. At each time-frequency cell, a Wilcoxon signed-rank z-statistic was computed on the pairwise differences across participants. Contiguous clusters of cells exceeding a z-threshold of a cluster-defining $\alpha = 0.10$ (two-tailed) were identified whereby cells were considered adjacent only if they shared an edge in the frequency-time plane

(horizontally or vertically), not diagonally. The cluster statistic was defined as the sum of z-values within each cluster. A null distribution of maximum cluster statistics was generated over 5,000 permutations. Clusters were considered significant if their statistic exceeded the 95th percentile of this null distribution ($\alpha = 0.05$), and a minimum cluster size of 2 time-frequency bins was required.

Finally, EEG power was averaged within each frequency band per ROI, yielding within-condition power indices for each of the three conditions (High-RM, Low-RM, Silence). Pairwise difference scores were also computed by subtraction (High-RM vs. Silence, Low-RM vs. Silence, High-RM vs. Low-RM) to index condition-related changes in band power. Spearman correlations were conducted between these EEG indices and scores on individual difference measures (i.e., DASS-21, BMMR, BFI-2-S). To control the false discovery rate (FDR), *p*-values were corrected using the Benjamini-Hochberg procedure separately within each questionnaire family with a corrected significant threshold of $p < .05$.

2.6.2. Exploratory three-dimensional cluster-based permutation analyses

To complement the *a priori* ROI analyses, exploratory cluster-based permutation tests were conducted across all channels simultaneously within each frequency band (delta, theta, alpha, beta) using the same epoch segment of 10 - 170 seconds. Analyses were implemented using the FieldTrip toolbox, revision 20250106 [74], with permutation testing carried out using FieldTrip's Monte Carlo method. Data were aggregated into non-overlapping 10-second time bins prior to analyses. The channel neighbour structure was constructed using the 'distance' method within FieldTrip's "ft_prepare_neighbours" function. At each channel-frequency-time point, a paired-samples *t*-statistic was computed, with condition labels permuted within each participant independently. Contiguous clusters of points exceeding a cluster-defining threshold of $\alpha = 0.05$ (two-tailed) were identified, with the requirement that each channel in a cluster must have at least two spatial neighbours also exceeding the threshold. The cluster statistic was defined as the sum of *t*-values within each cluster. A null distribution of maximum cluster statistics was generated over 5,000 permutations. Clusters were considered significant if their statistic exceeded the 95th percentile of this null distribution (test $\alpha = 0.05$).

2.7. Behavioral data analysis

2.7.1. Emotion ratings after music listening conditions

To characterize the affective responses elicited during the music conditions, descriptive statistics of discrete emotion ratings from the Geneva Music-Induced Affect Checklist (GEMIAC) [63] were first tabulated. For inferential analyses, arousal-valence ratings were selected as the primary outcome emotion measures given their direct correspondence with AffectMachine's affective parameter space.

Self-reported arousal and valence ratings were analyzed using linear mixed effects models (LME) fitted with the *lme4* package [75] in R version 4.5.2, with Satterthwaite-corrected degrees of freedom and *p*-values obtained via *lmerTest* [76]. For each emotion dimension, a single model was specified with time-point (pre-math, post-math, post-condition), condition (Silence, High-

relaxation music, Low-relaxation music), and their interaction as fixed effects. Two nuisance covariates were additionally included: music presentation order ("music_order"; whether High- or Low-relaxation music was presented first) to control for potential carryover or order effects arising from the counterbalanced design, and block number ("block"; 1, 2, 3) to account for the nested temporal structure of the experiment whereby the post-condition rating of each block serves as the pre-math baseline of the subsequent block. The resulting model was:

$$\text{rating} \sim \text{time-point} * \text{condition} + \text{music_order} + \text{block} + (1 \mid \text{participant})$$

Random intercepts were included for participant to account for individual differences in baseline ratings. We initially specified a maximal random effects structure including by-participant random slopes for time-point; however, this model was singular. Removing the intercept-slope correlation did not resolve convergence, and we therefore adopted a random intercept-only structure, which converged without singularity. This simplification is consistent with recommendations for parsimonious random effects structures when more complex specifications are not supported by the data [77].

Planned pairwise contrasts were extracted using the *emmeans* package v2.0.3 [78] to address three planned research questions: (1) whether arousal and valence changed significantly from pre- to post-math (stress induction manipulation check), (2) whether arousal-valence ratings recovered from post-math to post-condition (recovery effect), and (3) whether post-condition ratings returned to pre-math baseline levels. All contrasts were Bonferroni-corrected for three comparisons per condition, with statistical significance set at $\alpha = .05$.

2.7.2. Associations between Arousal-Valence ratings and questionnaires for individual differences

To quantify the degree of change in self-reported levels of arousal or valence elicited by the silence/music conditions, a change score was calculated as:

$$\Delta \text{ valence} = \text{valence post-condition} - \text{valence pre-condition}$$

$$\Delta \text{ arousal} = \text{arousal post-condition} - \text{arousal pre-condition}$$

whereby successful recovery from stress is represented by a positive Δ valence (i.e., increased valence) and negative Δ arousal (i.e., reduced arousal levels). Thereafter, simple linear regressions were conducted to examine whether individual difference measures (DASS-21 subscales, B-MMR subscales, BFI-2-S domains) predicted Δ arousal and with Δ valence scores in the Low- and High-relaxation music conditions. To correct for multiple comparisons, false discovery rate correction (FDR; Benjamini-Hochberg) was applied separately within each questionnaire family.

3. Results

3.1. Participant Demographics

One participant did not respond to questions on musical background specifically, resulting in $n = 26$ for the following descriptive statistics; all other variables had complete data. Of these respondents, nine participants (34.6%) reported no experience playing a musical instrument. Eight participants (30.8%) reported to be currently playing an instrument ($M_{year} = 20.75$ years, $SD_{year} = 12.18$). The remaining nine participants (34.6%) reported prior instrument-playing experience but indicated they had stopped playing, on average $M_{year} = 6.33$ years ($SD_{year} = 10.70$) prior to the study. The most commonly reported music listening frequency was 8-14 hours per week ($n = 8$, 30.8%) (Table 1).

For the DASS-21 subscales, mean scores ranged from 29.00 to 33.30, with the highest mean observed for Anxiety, followed by Depression and Stress. For the B-MMR subscales, mean scores ranged from 7.96 to 12.30, with the lowest mean on the Discharge strategy and highest on the Strong Sensation strategy. Lastly, for BFI-2-S subscales, mean scores ranged from 2.68 to 3.54, with the highest mean on Agreeableness and lowest on Extraversion (Fig. 3, Table 1).

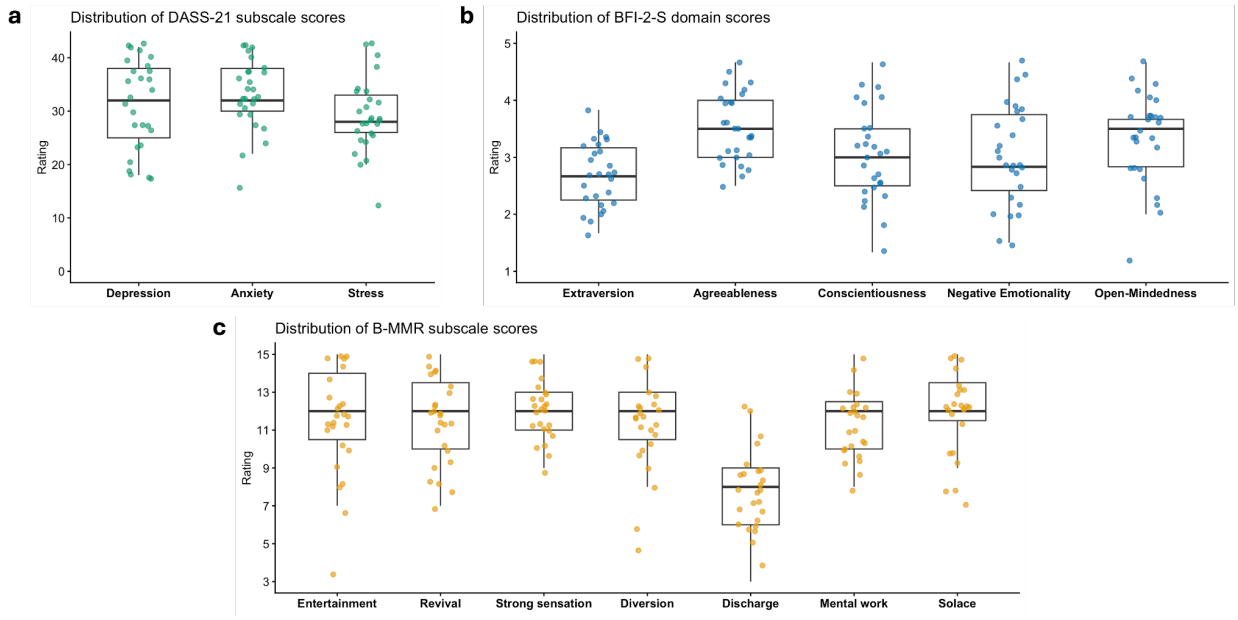


Figure 3. Boxplots of questionnaires for individual differences.

Table 1. Participant demographics

	N	(%)	M	SD
Age			31.33	9.65
Gender	10M 16F 1 Other			
Musical background and experience*				
Hours spent listening to music per week				
< 1 hour	1	4%		
1 - 3 hours	5	19%		
4 - 7 hours	6	23%		
8 - 14 hours	8	31%		
> 15 hours	6	23%		
Experience playing an instrument				
No	9	35%		
Yes (currently playing; years of playing)	8	31%	20.75	12.18
Yes (prior experience but not currently playing; years since last played)	9	35%	6.33	10.70
DASS-21				
Depression			31.40	8.37
Extremely severe	19	70%		
Severe	3	11%		
Moderate	5	19%		
Anxiety			33.30	6.47
Extremely severe	26	96%		
Severe	1	4%		
Stress			29.00	6.85
Extremely severe	7	26%		
Severe	14	52%		
Moderate	5	19%		
Normal	1	4%		
BMMR				
Entertainment (ENT)			11.60	2.90
Revival (REV)			11.60	2.34
Strong sensation (SS)			12.30	1.75
Diversion (DIV)			11.60	2.58
Discharge (DIS)			7.96	2.58
Mental work (MW)			11.60	2.02

Solace (SOL)	12.10	2.29
BFI-2-S		
Extraversion	2.68	0.56
Agreeableness	3.54	0.61
Conscientiousness	3.04	0.81
Negative Emotionality	3.02	0.87
Open-Mindedness	3.33	0.79

* One participant did not respond to questions on musical background specifically, resulting in $n = 26$ for the relevant descriptive statistics; all other variables reported here had complete data. Percentages are rounded to the nearest whole number.

3.2. Behavioral results

3.2.1. Manipulation check: Adaptive Mental Arithmetic Task (AMAT)

On a continuous scale ranging from 0 (*not at all*) to 100 (*agree completely*), participants moderately agreed that they are good at maths ($M = 41.1$, $SD = 22.6$). Regarding the AMAT task, they largely agreed that it was stressful ($M = 74.7$, $SD = 23.5$) and wished for additional time to answer the questions ($M = 91.1$, $SD = 16.2$). Consistent with these reports, participants largely disagreed that the questions were easy ($M = 24.9$, $SD = 19.0$) and expressed low satisfaction with their task performance ($M = 27.3$, $SD = 24.3$; Fig. 4).

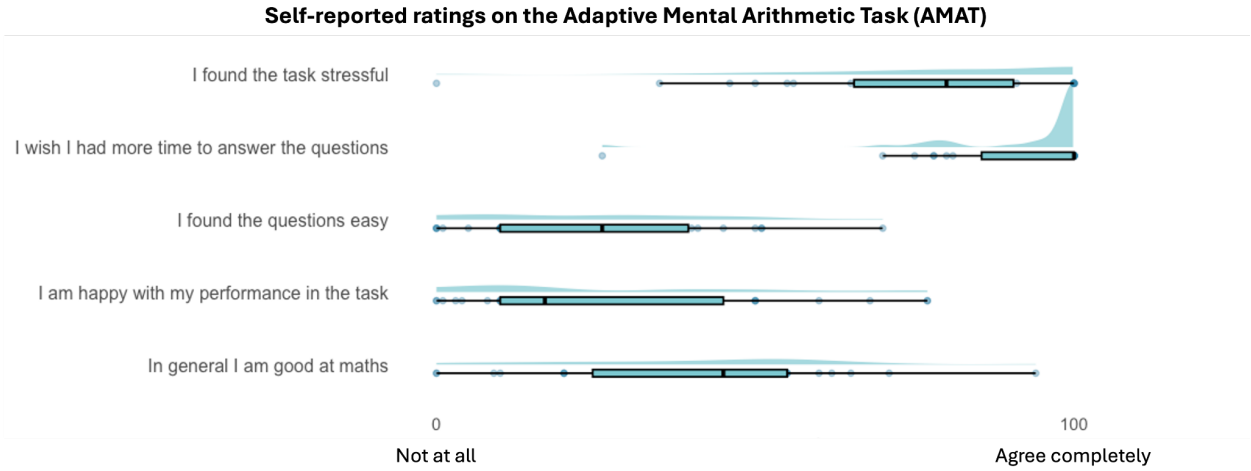


Figure 4. Self-reported ratings on the Adaptive Mental Arithmetic Task

3.2.2. Emotion ratings after music listening conditions

Both High-RM and Low-RM evoked positive affect, with 'relaxed' receiving the highest self-reported rating (**High-RM:** $M = 3.56$, $SD = 0.75$; **Low-RM:** $M = 3.58$, $SD = 0.83$) and 'agitated' the lowest (**High-RM:** $M = 1.72$, $SD = 0.65$; **Low-RM:** $M = 1.89$, $SD = 0.79$; see Supplementary Fig. S1 online).

Arousal ratings. LME analyses with self-reported arousal ratings as outcome revealed a significant main effect of time-point ($F(2, 207) = 23.65$, $p < .001$), indicating that arousal ratings varied meaningfully across the three measurement points (i.e., pre-math, post-math, post-condition). Neither the main effect of condition nor the timepoint*condition interaction reached significance at $p < .05$, suggesting that the overall arousal trajectory did not differ significantly among conditions. Nuisance covariates of music presentation order and block number were both not significant ($p > .05$), confirming the absence of order or fatigue effects (Table 2). Planned pairwise contrasts with Bonferroni-correction showed that, as a manipulation check for the arithmetic task eliciting stress, arousal increased significantly from pre- to post-math across all three conditions (silence: $t(207) = 3.80$, $p = .001$; low-relaxation: $t(207) = 3.78$, $p = .001$; high-relaxation: $t(207) = 3.60$, $p = .001$), confirming that the arithmetic task successfully elevated arousal. For the recovery phase, arousal decreased significantly from post-math to post-condition in the silence condition ($t(207) = 4.40$, $p < .001$) and High-relaxation music ($t(207) = 1.57$, $p = .352$) condition, but not in the Low-relaxation music condition ($t(207) = 3.19$, $p = .005$). Post-condition arousal did not differ significantly from pre-math baseline in any condition (all $p > .13$), indicating that participants returned to baseline arousal levels after the silence/music conditions (Fig. 5a, Table 3).

Valence ratings. LME analyses with self-reported valence ratings as outcome revealed significant main effects of timepoint ($F(2, 207) = 21.16$, $p < .001$) and block ($F(1, 207) = 5.16$, $p = .024$), but not condition ($F(2, 207) = 2.62$, $p = .075$). The timepoint*condition interaction was significant ($F(4, 207) = 5.66$, $p < .00$), indicating that the trajectory of valence across measurement points differed among conditions. Lastly, music presentation order was not significant ($p = .713$), confirming the absence of order effects (Table 2). Planned pairwise contrasts with Bonferroni-correction revealed divergent patterns across conditions. In the silence condition, valence decreased significantly from pre- to post-math ($t(207) = -5.26$, $p < .001$), did not recover significantly after the silence period ($t(207) = 1.29$, $p = .600$), but ended up significantly lower than pre-math baseline at post-condition ($t(207) = -3.97$, $p < .001$). In the High-relaxation music condition, valence did not decrease significantly after math ($t(207) = -1.84$, $p = .203$), but increased significantly after music ($t(207) = 4.11$, $p < .001$), with a trend towards ending above pre-math baseline ($t(207) = 2.28$, $p = .071$). In the Low-relaxation music condition, valence decreased significantly after math ($t(207) = -2.88$, $p = .013$), and recovered significantly after music ($t(207) = 4.13$, $p < .001$), and showed a clean return to pre-math baseline levels ($t(207) = 1.26$, $p = .630$). Together, these findings indicate that both music conditions effectively restored valence following the arithmetic task, whereas silence did not produce a comparable recovery (Fig. 5b, Table 3).

Emotion ratings summary. Both High-RM and Low-RM evoked positive felt affect, with 'relaxed' receiving the highest self-reported rating across both conditions. Considering arousal and valence trajectories jointly, High-RM emerged as the only condition producing recovery in both

dimensions simultaneously: a significant reduction in arousal and a significant increase in valence from post-math to post-condition. Low-RM restored valence comparably but did not significantly lower arousal, while silence reduced arousal but did not restore valence to baseline. Interpretation of the valence findings warrants caution, as valence ratings showed a significant effect of block ($F(1, 207) = 5.16$, $p = .024$), indicating a systematic upward drift across the session that was independent of condition. However, the advantage of High-RM in simultaneously recovering both arousal and valence is unlikely to be fully attributable to this block trend, given that all three conditions were counterbalanced across blocks and the block covariate was included in the model. Taken together, these findings suggest that High-RM may offer the most comprehensive affective recovery following stress.

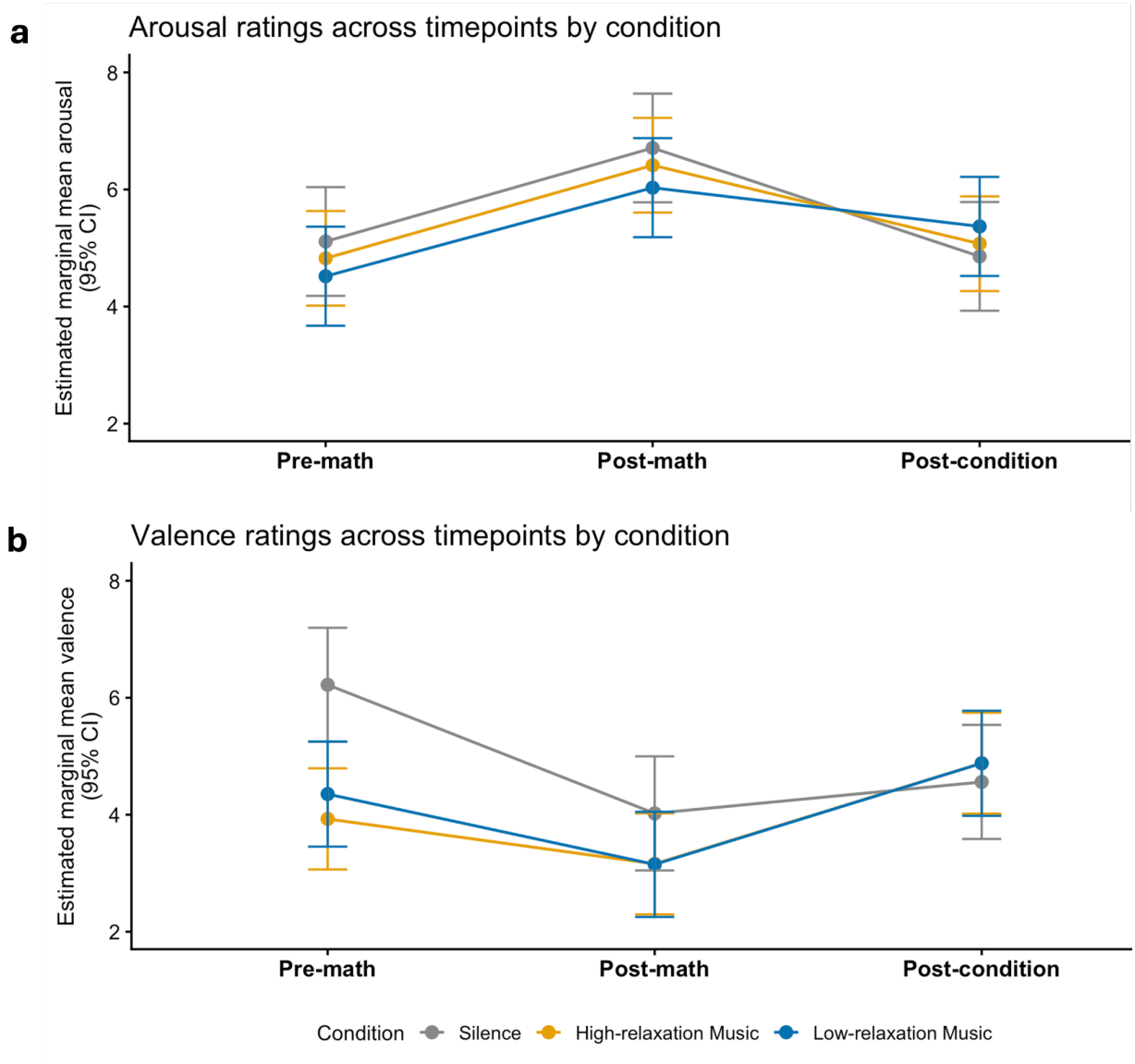


Figure 5. (a) Arousal ratings at the timepoints before math task, after math task, and after silence/music listening conditions. (b) Valence ratings at the timepoints before math task, after

math task, and after silence/music listening conditions. *Note.* Post-math scores for blocks 2 and 3 were the post-condition scores for blocks 1 and 2 respectively.

Table 2. Linear mixed-effects models: Type III ANOVA tables for self-reported arousal and valence ratings

Effect	**Sum Sq**	**Mean Sq**	Num*df*	Den*df*	*F*	*p*†
Arousal						
Timepoint	113.043	56.522	2	207	23.65	< .001 ***
Condition	0.903	0.452	2	207	0.19	.828
Music order	0.161	0.161	1	25	0.07	.797
Block	2.216	2.216	1	207	0.93	.337
Timepoint × Condition	11.967	2.992	4	207	1.25	.290
Valence						
Timepoint	100.009	50.005	2	207	21.16	< .001 ***
Condition	12.395	6.197	2	207	2.62	.075
Music order	0.326	0.326	1	25	0.14	.713
Block	12.184	12.184	1	207	5.16	.024 *
Timepoint × Condition	53.529	13.382	4	207	5.66	< .001 ***

Note. Model formula (fit separately for each outcome): rating ~ timepoint × condition + music_order + block + (1 | participant). Optimizer: BOBYQA. Satterthwaite degrees of freedom. Num df = numerator degrees of freedom; Den df = denominator degrees of freedom. † Uncorrected p-values; planned contrasts are reported in Table 3 with Bonferroni correction. * $p < .05$; *** $p < .001$.

Table 3. Bonferroni-corrected planned pairwise contrasts for self-reported arousal and valence ratings across timepoints and conditions

Contrast	Condition	Estimate	SE	*t*	*df*	*p*Bonf
Arousal						
Post-math vs. pre-math	*Silence*	1.598	0.421	3.80	207	.001 **
	High-relaxation	1.590	0.421	3.78	207	.001 **
	Low-relaxation	1.513	0.421	3.60	207	.001 **
Post-condition vs. post-math	*Silence*	−1.852	0.421	−4.40	207	< .001 ***
	High-relaxation	−1.341	0.421	−3.19	207	.005 **
	Low-relaxation	−0.662	0.421	−1.57	207	.352
Post-condition vs. pre-math	*Silence*	−0.254	0.421	−0.60	207	1.000
	High-relaxation	0.249	0.421	0.59	207	1.000
	Low-relaxation	0.851	0.421	2.02	207	.133
Valence						
Post-math vs. pre-math	*Silence*	−2.199	0.418	−5.26	207	< .001 ***
	High-relaxation	−0.769	0.418	−1.84	207	.203
	Low-relaxation	−1.203	0.418	−2.88	207	.013 *
Post-condition vs. post-math	*Silence*	0.538	0.418	1.29	207	.600
	High-relaxation	1.721	0.418	4.11	207	< .001 ***
	Low-relaxation	1.729	0.418	4.13	207	< .001 ***
Post-condition vs. pre-math	*Silence*	−1.661	0.418	−3.97	207	< .001 ***

High-relaxation	0.953	0.418	2.28	207	.071
Low-relaxation	0.526	0.418	1.26	207	.630

Note. Contrasts were computed from the estimated marginal means of the linear mixed-effects models (Table 2) using the emmeans package, and are expressed as the later timepoint minus the earlier timepoint. For arousal, a positive estimate indicates an increase in arousal and a negative estimate indicates a decrease in arousal. For valence, a positive estimate indicates an increase in valence (more positive affect) and a negative estimate indicates a decrease in valence (more negative affect). *p*-values are Bonferroni-corrected for three contrasts per condition within each outcome. SE = standard error; df = degrees of freedom. * $p < .05$; ** $p < .01$; *** $p < .001$.

3.2.3. Associations between Arousal-Valence ratings and questionnaires for individual differences

After FDR-correction, only the DASS-21 Stress subscale remained significant, negatively predicting Δ valence in both the High-relaxation ($b = -0.170$, $SE = 0.054$, $t = -3.15$, $p = .025$) and Low-relaxation ($b = -0.215$, $SE = 0.054$, $t = -3.97$, $p = .006$) music conditions, indicating that participants who reported higher baseline stress showed smaller valence recovery in both music conditions (Fig. 6). No B-MMR or BFI-2-S subscales significantly predicted Δ arousal or Δ valence in either condition after correction (see Supplementary Table S2 online for full results).

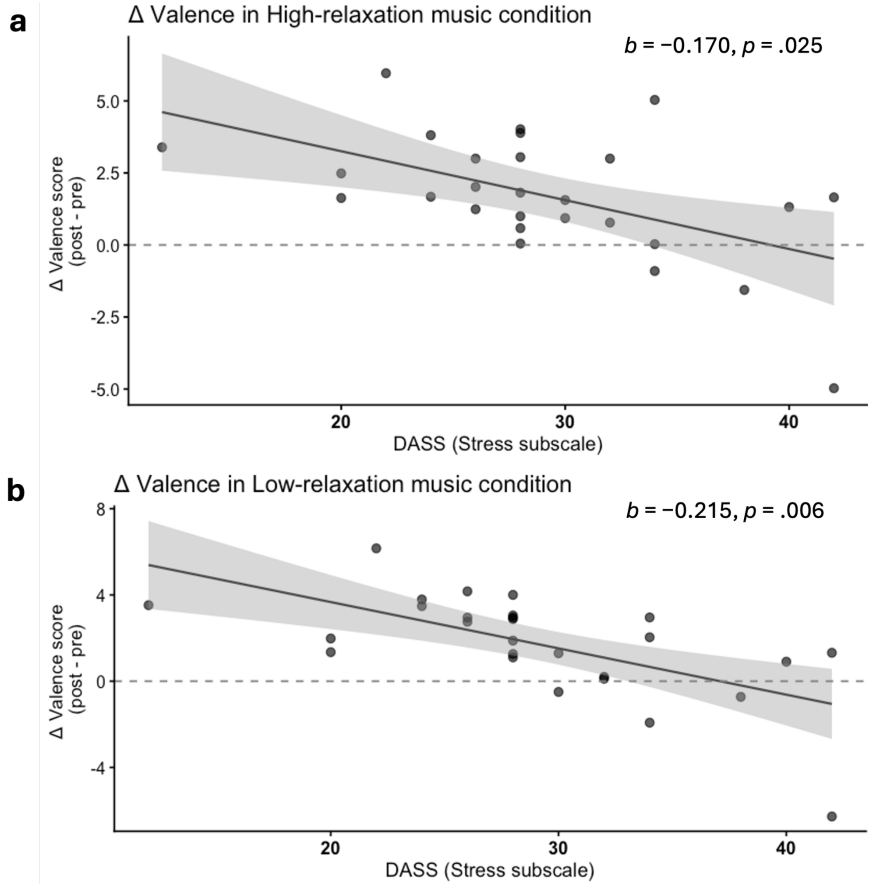


Figure 6. Relationship between DASS-21 Stress subscale scores and valence change scores (Δ valence) in the (a) High-relaxation music condition and (b) Low-relaxation music condition. Note. Δ valence was calculated as valence rating at post-condition minus valence rating at pre-condition, such that positive values indicate a net increase in valence and negative values indicate a net decrease in valence. Solid lines represent the fitted linear regression, and shaded bands represent the 95% confidence interval. The dashed horizontal line marks Δ valence = 0 (no net change from pre-condition).

3.3. *A priori* ROI-level time-frequency analyses

Cluster-based permutation tests using Wilcoxon signed-rank z-statistic revealed significant differences for both music conditions compared to silence (i.e., High-relaxation music (High-RM) > Silence, Low-relaxation music (Low-RM) > Silence), but not between the music conditions (i.e., High-RM > Low-RM). Wilcoxon signed-rank test was chosen as one or more group-level comparison showed that more than 50% of time-frequency cells violated normality assumptions (see Supplementary Fig. S2-4 online).

Relative to the silence condition, suppressed delta-band (2-4 Hz) activity was observed in the frontal ROI in both High-RM and Low-RM conditions. Low-RM additionally exhibited delta suppression in the parietal ROI, an effect not observed for High-RM. No significant effect was observed in the central ROI across any comparison (Fig. 7; Table 4).

Table 4. Cluster statistics representing significant differences between contrasts in delta frequency band

	Delta [2-4 Hz]				
Clusters reflecting significant differences	Region	Time range (s)	cluster size	peak *z*	*p*
High-Relaxation music > Silence					
Cluster 1	Frontal	37.5 to 77.5	14	-2.84	.025
Low-Relaxation music > Silence					
Cluster 1	Frontal	37.5 to 87.5	19	-2.93	.017
Cluster 2	Frontal	97.5 to 142.5	13	-3.03	.043
Cluster 3	Parietal	42.5 to 87.5	13	-3.1	.018
Cluster 3	Parietal	97.5 to 167.5	16	-3.6	.007

Note. All contrasts with significant differences resulted in negative clusters (i.e., power suppression during music listening as compared to silence). No significant differences were observed in alpha, theta, or beta frequencies.

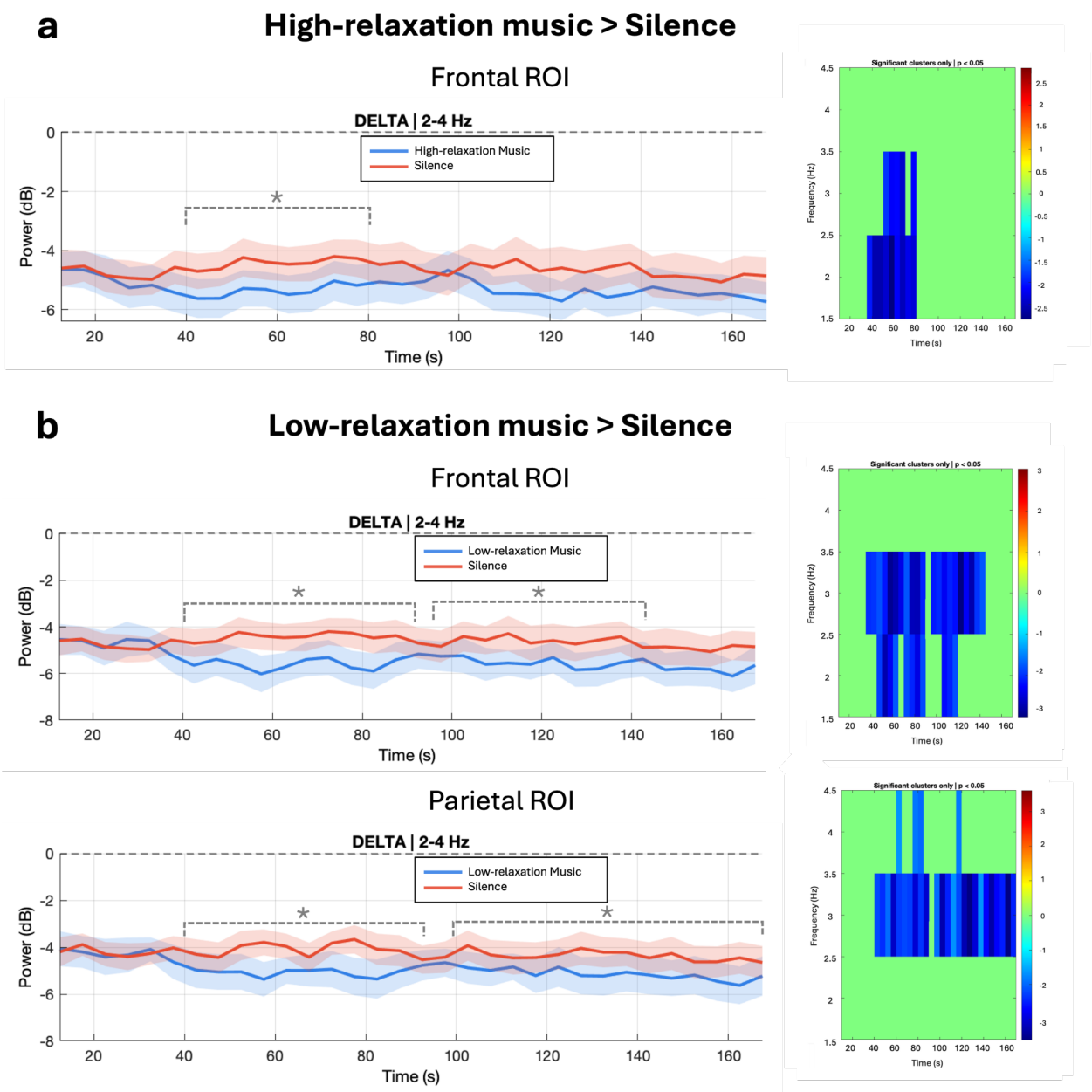


Figure 7. (a) Significant differences in frontal delta between High-relaxation music (High-RM) > Silence. (b) Significant differences in frontal and parietal delta in Low-relaxation music (Low-RM) > Silence. Note: High-RM > Low-RM did not yield any significant differences. *$p < .05$.

3.4. Correlation with questionnaires for individual differences

Spearman correlations between condition-related changes in band power (i.e., High-RM - Silence, Low-RM - Silence, High-RM - Low-RM) and scores on individual difference measures did not yield any significant findings after FDR correction at $p < .05$. However, significant positive associations emerged between DASS-21 scores and the averaged band power of each condition: in the High-RM condition, frontal and parietal delta activities are positively correlated with all

three subscales of DASS, and the Anxiety subscale is further positively correlated with central delta activity and parietal theta activity. In the Low-RM condition, the Anxiety subscale was positively correlated with frontal delta activity. Lastly, in the silence condition, the Anxiety subscale is positively correlated with delta activity in the frontal, central, and parietal regions (Fig. 8; see Supplementary Table S3 online).

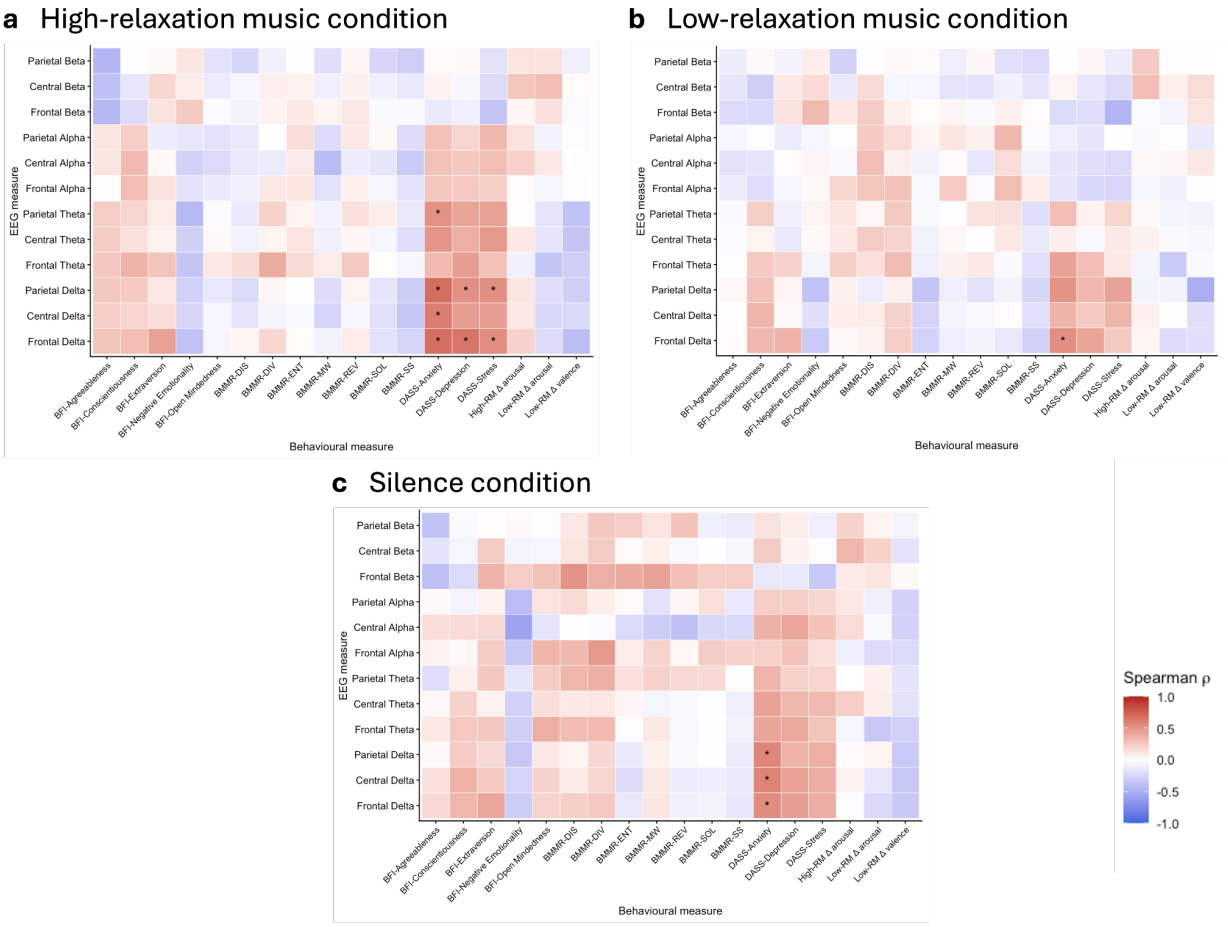


Figure 8. Spearman correlations between scores on individual differences on DASS, BMMR, BFI-2-S and EEG indices in the (a) High-relaxation music condition, (b) Low-relaxation music condition, and (c) silence condition. * indicates FDR-corrected $p < .05$

3.5. Exploratory three-dimensional cluster-based permutation analyses

3D cluster-based permutation analyses similarly showed significant differences for both music conditions compared to silence (i.e., High-RM > Silence, Low-RM > Silence), but not between the music conditions (i.e., High-RM > Low-RM). A significant difference in the alpha frequency band was found between High-RM and silence from 55 - 105 sec. This difference was most pronounced over central, parietal and occipital electrodes (Fig. 9a). In contrast, significant differences in the delta frequency band were found between Low-RM and silence. The first cluster spanned the duration of 45 - 85 sec, consisting of electrodes from bilateral frontal and parietal

regions. The second cluster spanned the duration of 125 - 175 sec, consisting of electrodes biased to the right frontal and bilateral parietal regions (Fig. 9b).

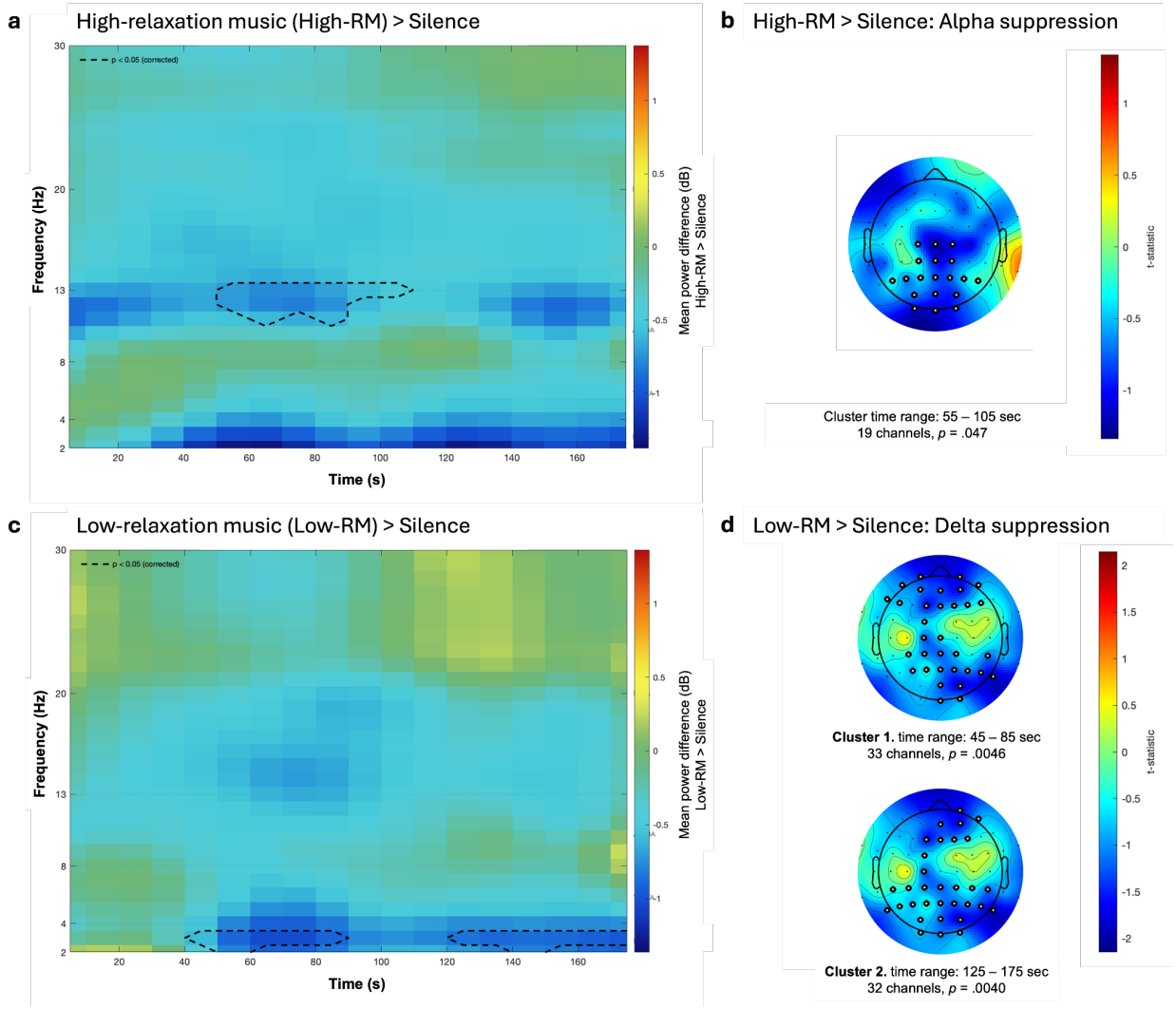


Figure 9. (a) High-relaxation music > Silence: group mean power difference. (b) Topographic map of significant difference in High-relaxation music > Silence. (c) Low-relaxation music > Silence: group mean power difference. (d) Topographic map of significant differences in Low-relaxation music > Silence.

4. Discussion

The current study investigated the electrophysiological correlates of stress recovery using generated affective music which differed in the degree of relaxation content in a sample of adults with elevated anxiety. Results showed that after taking into account the listener's preferred music genre (classical or pop), only High-relaxation music (High-RM) produced recovery in both arousal and valence dimensions simultaneously, while Low-relaxation music (Low-RM) restored valence

comparably but failed to reduce arousal significantly. Compared to both music conditions, the silence condition reduced arousal but did not restore valence to baseline.

The efficacy of High-RM in regulating both arousal and valence towards the desired relaxation state can be understood through the combined logic of the iso principle [47] and the mood congruence hypothesis (MCH) [79]. In the current study, High-RM was designed to express music of a higher degree of relaxation than Low-RM, with affective qualities first calibrated near participants' post-stress state and shifted 30% toward positive-valence and low-arousal, rather than generating music reflecting a fully relaxed state. This partial-match design draws on the logic underlying the iso principle, in that the music does not immediately impose a fully relaxed target state but instead offers a state more proximal to the listener's own emotion state. This proximity, in turn, may explain why the music was successfully engaged with: according to the MCH, information or stimuli that are affectively congruent with one's current state are processed more readily and preferentially [79], and music that is mood-congruent has been shown to be more strongly engaged with and preferred over incongruent music [52]. Taken together, High-RM may have been effective precisely because it was congruent enough with participants' post-stress affective state to be readily engaged with (mood congruence), while still being sufficiently shifted toward a calmer, more positive target to elicit genuine recovery in both arousal and valence (embodiment of the iso principle).

Interestingly, our findings point toward the efficacy of Low-RM as well in regulating valence, although not arousal. This is somewhat surprising, as music is generally more effective at reducing arousal than shifting valence [46, 80-81]. This positive effect on shifting valence in both music conditions is reminiscent of studies that compared the effects of iso- versus compensatory playlists on emotion regulation in healthy adults and found no differences between the effectiveness of both types of playlists in regulating emotions compared to a control condition [49, 54]. Another notable observation is the decrease in arousal observed in the silence condition, which may be attributed to either active self-regulation (e.g., controlled breathing, reappraisal; [82]) or passive physiological recovery, which does not require deliberate intervention [83].

4.1. Fronto-parietal suppression in music conditions relative to silence

The *a priori* ROI-level time-frequency analyses revealed that both music conditions suppressed frontal delta activity relative to silence. Increased delta power, particularly over frontal regions, has been reported across depression and anxiety [13]. This account is further supported by the positive correlations between the Anxiety subscale of DASS-21 and frontal delta power of each condition observed in the current study. To the extent that frontal delta indexes trait-level anxious distress, this convergence between existing literature and the present within-sample correlations lends support to interpreting the observed suppression of frontal delta during music listening, relative to silence, as reflecting a shift away from the distress-related neural state induced by the preceding arithmetic stressor. This interpretation is broadly consistent with the behavioral finding that both music conditions produced comparable stress recovery on self-reported valence.

Interestingly, Low-RM produced an additional delta suppression in the parietal region. Delta oscillations have been proposed to index attention to internally directed processing, with delta increases reflecting inhibition of external sensory interference during effortful internal concentration [84]. Accordingly, it is plausible that this parietal delta suppression during Low-RM may reflect a shift away from internally-directed processing toward external, stimulus-driven attention, given that Low-RM was generated at levels close to the post-stressor affective state.

Exploratory 3D cluster-based permutation analyses yielded a broadly similar pattern of results compared with the *a priori* ROI-level time-frequency analyses. Delta suppression was also observed in Low-RM relative to silence, with contributing electrodes from frontal and posterior regions. This is congruent with the finding of fronto-parietal delta suppression from the ROI analyses, with the exploratory analyses showing that the cluster included more neighboring electrodes compared to those defined in the ROI analyses.

4.2. Parieto-occipital alpha suppression in High-relaxation music relative to silence

Interestingly, parieto-occipital alpha suppression emerged in the comparison of High-RM relative to silence in the exploratory cluster-based permutation analyses, but not in the *a priori* ROI-level analyses. Of note, the exploratory analyses considered all 61 channels simultaneously within each frequency band and identified clusters based on spatial adjacency across the full scalp [73]. This approach can thus detect contiguous electrodes irrespective of pre-defined ROI boundaries. Consistent with this account, the parieto-occipital cluster identified to be different in the High-RM versus silence comparison included electrodes extending beyond the boundaries of the parietal ROI defined in the *a priori* analysis. Given the exploratory nature of this analysis, this effect should be interpreted with appropriate caution.

The observed alpha suppression may be understood through the lens of attentional engagement. Alpha-band suppression has long been considered a neural marker of attentional engagement with the external environment, with greater suppression reflecting increased attentional engagement towards the external environment [85, 86]. Under this account, the parieto-occipital alpha suppression observed during High-RM may reflect heightened attentional engagement with the music itself. This is consistent with the proposal that High-RM, being calibrated to be proximal to but not fully divergent from participants' post-stress affective state, facilitated greater engagement with the music (i.e., mood congruence hypothesis; [52, 79]), an observation not found in Low-RM.

4.3. Individual differences: Stress is associated with valence recovery while trait anxiety is associated with delta power across conditions

Findings from the current study showed that the effect of increase in valence for both music conditions was larger for individuals with a lower score on the Stress subscale of DASS-21. However, the Stress subscale of DASS-21 was not extensively associated with electrophysiological markers within condition and condition-related changes. Instead, the strength and topographical extent of the association between DASS-21 scores and delta power varied

systematically by condition: In silence, higher scores on the Anxiety subscale were associated with higher delta power across frontal, central, and parietal regions; in Low-RM, a positive correlation between the Anxiety subscale and delta power was observed only in the frontal region; in High-RM, however, associations between behavioral measures and EEG were both broader across regions and extended to all three DASS-21 subscales, including an additional link between Anxiety and parietal theta, which were all in a positive direction.

One plausible explanation draws on the iso principle and compensatory listening frameworks. Given that High-RM was calibrated to be more affectively discrepant from participants' post-stress state than Low-RM [47, 48], engaging with it may have required more active regulatory work for individuals with elevated anxiety, who have been shown to rely more heavily on compensatory, mismatch-driven regulation strategies than mood-matched ones [50]. This would be consistent with attentional control theory, which proposes that individuals with elevated anxiety recruit additional processing resources to sustain goal-directed engagement, particularly under conditions of greater affective or attentional demand [5]. Accordingly, a broader and more effortful neural engagement with High-RM among more anxious participants would therefore be expected. Alternatively, this broader pattern may simply reflect a more general association between trait negative affect and neural markers of engagement or effort during exposure to strongly positively valenced stimuli, a possibility that cannot be disambiguated with the current data.

While interesting patterns of correlations emerged between DASS-21 subscales and within-condition EEG power, no condition-related change in band power was significantly associated with any individual difference measure after FDR correction. This suggests that the degree of neural change elicited by music relative to silence was not reliably moderated by trait anxiety, depression, stress, habitual music-based mood regulation strategies (as measured by B-MMR), or personality (as measured by BFI-2-S), in this sample of individuals with elevated anxiety. This null pattern is notable given that trait-level constructs such as Negative Emotionality/Neuroticism and maladaptive music-regulation strategies (e.g., Discharge) have previously been linked to affect-related neural responses to music [87].

Taken together, these findings suggest a divergence between the subjective and neural correlates of individual differences in stress recovery: whereas the DASS-21 Stress subscale moderated the magnitude of behavioral valence recovery, no individual difference measure moderated condition-related neural change. This divergence should be interpreted with appropriate caution, but raises the possibility that self-reported affective recovery and the neural markers examined here may not be capturing fully overlapping aspects of the stress-recovery process.

4.4. Limitations & future directions

While this is one of the first studies that examined stress recovery with generated affective music of differing levels of relaxation, it is not without limitations. In this study, we did not explicitly measure stress-level after each block of arithmetic task, but instead measured negative affect (i.e., self-reported valence-arousal ratings). It is plausible that the lack of effect in reducing

arousal level could be attributed to the intensity of the arithmetic task. Studies have reported that music may only be effective in reducing stress in the context of a mild stressor as opposed to a strong stressor [32, 80]. Further, we employed a passive control condition whereby the participants simply remained silent after the arithmetic task. Thus, for the comparisons between music conditions against silence, we cannot effectively rule out alternative explanations such as general auditory engagement or attentional capture by the music stimuli. In addition, there may be a potential confounding effect of music listening for the next stress block. While this has been accounted for in our analyses, it could nonetheless affect the interpretations of stress-recovery in subsequent blocks. Lastly, the current sample scored highly across all DASS-21 by design, which could possibly limit the analyses of individual differences. Future research could examine a broader or more stratified range of anxiety severity to test whether this affected the individual difference analyses.

Results from this study nevertheless demonstrate an overall efficacy of using generated affective music for stress reduction and emotion regulation. We also demonstrate that listening to music of different degrees of relaxation could have an effect on self-reported felt emotions and involve different brain regions. There is thus potential in integrating generated affective music into neurofeedback systems to assist with emotion regulation in individuals with psychological distress. It would also be interesting to examine the system's efficacy with those diagnosed with mental illnesses. Lastly, future work should test the iso principle as compared to the compensatory approach in designing musical neurofeedback, in both those with clinical and non-clinically diagnosed anxiety. Overall, results from this study could inform future developments of musical neurofeedback and treatment interventions for those with undiagnosed anxiety/depression.

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Author Contributions Statement

G.H wrote the main manuscript text, refined data analyses pipelines, and prepared all visualizations. C.K performed preliminary data analyses and contributed to initial stages of manuscript preparation. L.S was involved in conceptualization of research design and led data collection. S.E and C.G supervised the methodological aspects of EEG data analyses and contributed to manuscript revision. K.A conceptualized the study, contributed to manuscript revision and editing, and supervised the research. All authors reviewed the manuscript.

ADDITIONAL INFORMATION

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Data availability

The dataset generated and analyzed in the current study is available from the corresponding author (K.A) upon reasonable request.

Competing interests

The authors declare no competing interests.

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